

A convergent series for OEIS A352117, and the reason it starts
at $n = 1$:

a proof of the Yanev–Kotesovec conjecture

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Abstract

OEIS A352117 has exponential generating function $1/\sqrt{2 - e^{2x}}$. In July 2026 Velin Yanev and Vaclav Kotesovec conjectured that $a(n) = \sum_{k \geq 1} 2^{n-3k-1/2} k^n \binom{2k}{k}$ for $n > 0$. We prove it. Forming the exponential generating function of the right-hand side and exchanging the order of summation turns the inner sum into the central binomial series $(1 - 4z)^{-1/2}$ evaluated at $z = e^{2x}/8$, and $1 - 4 \cdot \frac{e^{2x}}{8} = \frac{2 - e^{2x}}{2}$, so the $\sqrt{2}$ cancels. What is left over is the single term $k = 0$, excluded by the summation range, contributing the constant $1/\sqrt{2}$. That constant affects only the coefficient of x^0 , which is exactly why the conjecture is stated for $n > 0$ and false at $n = 0$: the discrepancy there is $1 - 1/\sqrt{2}$, and we verify that numerically. The interchange is justified by absolute convergence on $|x| < \frac{1}{2} \log 2$, the disc on which the generating function is analytic.

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1 The sequence and the conjecture

OEIS A352117 satisfies $a(n) = \sum_{k=0}^n 2^{n-k} (\prod_{j=0}^{k-1} (2j+1)) S(n, k)$ with $S(n, k)$ the Stirling numbers of the second kind. It has offset 0 and begins

$$a(0), \dots, a(7) = 1, 1, 5, 37, 377, 4921, 78365, 1473277, \dots$$

The entry records the exponential generating function

$$A(x) = \sum_{n \geq 0} a(n) \frac{x^n}{n!} = \frac{1}{\sqrt{2 - e^{2x}}}, \quad (1)$$

and carries the following comment:

Conjecture: $a(n) = \text{Sum}_{k \geq 1} 2^{(n-3k-1/2)} * k^n * \text{binomial}(2k, k)$ for $n > 0$. — *Velin Yanev*
and *Vaclav Kotesovec*, Jul 21 2026

As of the “Last modified” line on the live entry (22 July 2026, revision 43) the statement is still recorded as a conjecture, and no proof appears among the entry’s links, formulas or programs.

Write

$$S(n) := \sum_{k \geq 1} 2^{n-3k-1/2} k^n \binom{2k}{k} = \frac{2^n}{\sqrt{2}} \sum_{k \geq 1} \binom{2k}{k} \frac{k^n}{8^k}. \quad (2)$$

Since $\binom{2k}{k} \leq 4^k$ the terms are $O(k^n 2^{-k})$, so the series converges for every n . The claim to be proved is $S(n) = a(n)$ for all $n \geq 1$.

2 The interchange

Lemma 1. For $|z| < \frac{1}{4}$, $\sum_{k \geq 0} \binom{2k}{k} z^k = \frac{1}{\sqrt{1-4z}}$, absolutely.

Proof. The coefficient of z^k in $(1-4z)^{-1/2}$ is $\binom{-1/2}{k}(-4)^k = \binom{2k}{k}$, and $\binom{2k}{k} \leq 4^k$ gives radius $\frac{1}{4}$ with absolute convergence inside. $\square$

Lemma 2. For every real x with $|x| < \frac{1}{2} \log 2$,

$$\sum_{n \geq 0} S(n) \frac{x^n}{n!} = \frac{1}{\sqrt{2}} \sum_{k \geq 1} \binom{2k}{k} \frac{e^{2kx}}{8^k},$$

both sides converging absolutely.

Proof. From (2), $S(n)x^n/n! = \frac{1}{\sqrt{2}} \sum_{k \geq 1} \binom{2k}{k} 8^{-k} (2kx)^n/n!$, since $2^n k^n x^n = (2kx)^n$. Replacing x by $|x|$ makes every term non-negative, so by Tonelli's theorem the order of summation may be exchanged as soon as the total is finite. Summing over n first gives $\sum_n (2k|x|)^n/n! = e^{2k|x|}$, so the total is

$$\frac{1}{\sqrt{2}} \sum_{k \geq 1} \binom{2k}{k} \left(\frac{e^{2|x|}}{8}\right)^k,$$

finite by Lemma 1 exactly when $e^{2|x|}/8 < \frac{1}{4}$, that is $|x| < \frac{1}{2} \log 2$. $\square$

3 The theorem

Theorem 3. For every $n \geq 1$,

$$a(n) = \sum_{k \geq 1} 2^{n-3k-1/2} k^n \binom{2k}{k},$$

while for $n = 0$ the two sides differ by $1 - \frac{1}{\sqrt{2}}$. That is, the Yanev-Kotesovec conjecture on OEIS A352117 is true exactly as stated, including its restriction to $n > 0$.

Proof. Fix x with $|x| < \frac{1}{2} \log 2$. The sum in Lemma 2 runs from $k = 1$, so it is the full central binomial series of Lemma 1 at $z = e^{2x}/8$ with its $k = 0$ term, namely 1, removed:

$$\sum_{n \geq 0} S(n) \frac{x^n}{n!} = \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{1-4 \frac{e^{2x}}{8}}} - 1 \right).$$

The radicand is

$$1 - \frac{4e^{2x}}{8} = 1 - \frac{e^{2x}}{2} = \frac{2 - e^{2x}}{2},$$

so

$$\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{(2 - e^{2x})/2}} = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2 - e^{2x}}} = \frac{1}{\sqrt{2 - e^{2x}}} = A(x)$$

by (1). Hence

$$\sum_{n \geq 0} S(n) \frac{x^n}{n!} = A(x) - \frac{1}{\sqrt{2}}. \quad (3)$$

Both sides are analytic on $|x| < \frac{1}{2} \log 2$, so their Taylor coefficients agree. The constant $1/\sqrt{2}$ contributes only to the coefficient of x^0 . Therefore $S(n) = a(n)$ for every $n \geq 1$, while at $n = 0$ we get $S(0) = a(0) - 1/\sqrt{2} = 1 - 1/\sqrt{2}$. $\square$

Remark 4. The excluded index in the conjecture is not a numerical accident but the $k = 0$ term of the central binomial series. Deleting it removes the constant $1/\sqrt{2}$ from the generating function, and a constant can only disturb the coefficient of x^0 . This both explains why the conjecture must say $n > 0$ and predicts the exact size of the failure at $n = 0$, namely $1 - 1/\sqrt{2} = 0.2928932188\dots$, which is confirmed numerically below. As in the companion situation, the disc $|x| < \frac{1}{2} \log 2$ is exactly where $1/\sqrt{2 - e^{2x}}$ is analytic, its singularity lying at $e^{2x} = 2$.

4 Verification

All numbers here are outputs of a script written separately from this note.

The generating function identity (3) was checked symbolically: $\frac{1}{\sqrt{2}}((1 - 4e^{2x}/8)^{-1/2} - 1)$ and $1/\sqrt{2 - e^{2x}} - 1/\sqrt{2}$ simplify to the same expression.

The series was then summed independently in 40-digit arithmetic, truncated at $k = 4000$, with no reference to the generating function. For $n = 1, \dots, 6$ it returns

$$1, 5, 37, 377, 4921, 78365,$$

matching the entry's published terms to more than 12 decimal places in every case. At $n = 0$ it returns $0.29289321881345\dots$, while $a(0) = 1$; the difference is $0.707106781187\dots = 1/\sqrt{2}$, exactly as Theorem 3 predicts.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A352117, conjecture of V. Yanev and V. Kotesovec, 21 July 2026.
- [2] R. P. Stanley, *Enumerative Combinatorics*, Volume 2, Cambridge University Press, 1999, Chapter 5.
- [3] W. Rudin, *Principles of Mathematical Analysis*, 3rd ed., McGraw-Hill, 1976, Theorem 8.3.